

## Initial conditions in networks

If the unknown arbitrary constants are to be evaluated for particular solutions, other things must be known about the n/w described by the differential equation.

Evaluate the arbitrary const<sup>s</sup> that appear in the general solution of a differential eq<sup>n</sup>.  
Conditions existing at this instant are known as initial conditions (or) the initial state, conditions at

$t = \infty$ , known as final condition.

The time of throwing the switch is considered to be  $t=0$ ,  $t=0^-$  before throwing the switch  
Notation to distinguish two states of the n/w  $t=0^+$  after throwing the switch

At the reference time  $t=0$ , one or more switches operate <sup>switch</sup>

To differentiate b/w the time immediately before

(-) sign and immediately after the (+) sign operation of a switch.

Thus conditions existing just before the switch is

operated will be as  $i(0^-)$ ,  $V(0^-)$ , conditions after as  $i(0^+)$ ,  $V(0^+)$ .

Initial conditions in a n/w depend on the past history of the n/w prior to  $t=0^-$  and  $t=0^+$  (before) after switching. Past history will show up in the form

Exis<sup>t</sup> of capacitor v<sub>lgs</sub> & inductor currents. (Change in currents & v<sub>lgs</sub> of energy storing elements.



## Initial conditions in elements

The resistor: current and  $V_R$  are related by

Ohm's law,  $V = R I$

the current thro a resistor will change instantaneously if the  $V_R$  changes instantaneously.  $V_R$  will change instantaneously if current changes instantaneously.

Inductor: current cannot change instantaneously

closing a switch to connect an inductor to a source of energy will not cause current to flow at the initial instant, Inductor act as open ckt.

If a current  $I_0$  flows in the inductor at the instant switching takes place, the inductor can be acts as a current source of  $I_0$  amp

Capacitor:  $V_C$  cannot change instantaneously in a s/m of fixed capacitance.

If an uncharged capacitor is connected to an energy source, a current will flow instantaneously, the capacitor equivalent to short ckt. ie  $V = 0$

so that zero charge corresponds to zero  $V_C$  (short-ckt)  
with an initial charge, the capacitor is equivalent to  $V_C$  source

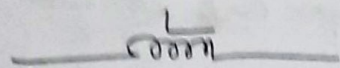
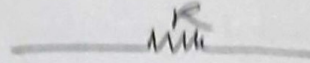
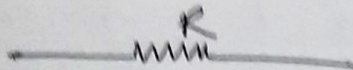
$$V_0 = \frac{q_0}{C}, q_0 \text{ is the initial charge.}$$



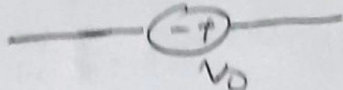
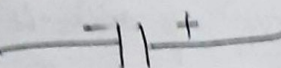
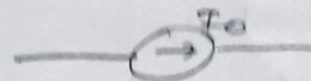
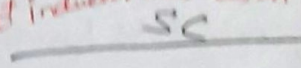
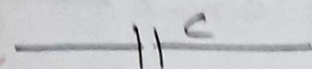
Equivalent form of the elements in terms of the initial condition of the element

Element  
(& initial condition)

Equivalent circuit  
at  $t=0^+$



Initially  $i(0^-) = 0$ ,  $i(0^+) = 0$   
means that at  $t=0$  inductor act as an open circuit



$$V_0 = \frac{q_0}{C}$$

$V(t) = V(0^-)$   
If  $V(0^-) = 0$   
then  $V(t) = 0$   
 $\therefore C$  acts as short circuit

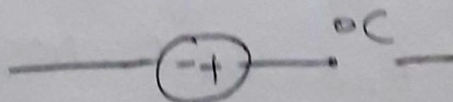
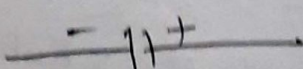
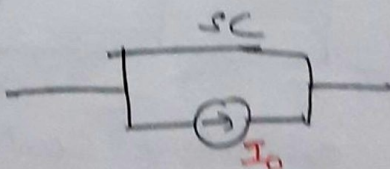
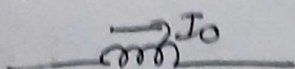
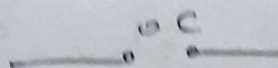
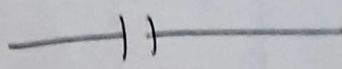
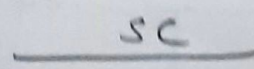
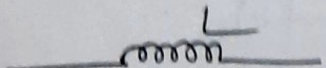
if  $i(0^-) = I_0$   
then  $i(t) = I_0$   
act as a current source

$i = C \frac{dV}{dt}$   
Under steady state  
 $\frac{dV}{dt} = 0$  condition

Equivalent form of the elements in terms of the final condition of the element

Element (Initial condition)

Equivalent circuit at  $t=\infty$   
(source causing constant steady state value)



$$V_0 = \frac{q_0}{C}$$

Steady state condition (final)

$V = L \frac{di}{dt}$   
Initial current  $\therefore \frac{di}{dt} = 0$   
 $\therefore V = 0$   
 $L$  act as short at  $t=\infty$   
when current flows

$$V_L = L \frac{di_L}{dt} \quad \text{and} \quad I_C = C \frac{dV_C}{dt}$$



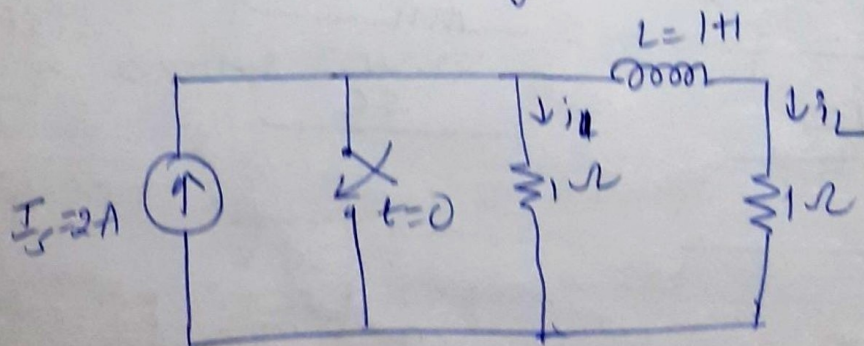
## Procedure for evaluating initial condition

There is no unique procedure to be followed in solving for initial conditions.

From the analysis, equivalent n/w for  $t=0^+$  may be constructed according to these rules.

- ① Replace all inductors with open ckt & with current generators having the value of current flowing at  $t=0^+$
- ② Replace all capacitors with short ckt & with a v/g source of value  $V_0 = \frac{Q_0}{C}$  if there is an initial charge.
- ③ Resistors are left into n/w without change.

Ex. Refer the ckt shown. Find  $i_1(0^+)$  and  $i_L(0^+)$ . The circuit is in steady state for  $t < 0$ .

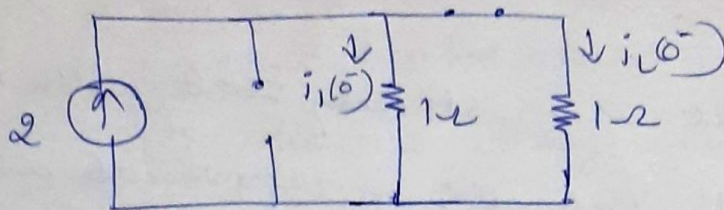


Solution: Symbol of switch implies that, it opens at  $t=0^-$  and closes at  $t=0^+$ . The ckt is in steady state with the switch open.



$\therefore$  At  $t = 0^-$ , inductor  $L$  act as short.

$\therefore$  At  $t = 0^-$



using current division rule,

$$i_L(0^-) = \frac{2 \times 1}{1+1} = 1A$$

Since current in an inductor cannot change instantaneously, we have

$$i_L(0^+) = i_L(0^-) = 1A$$

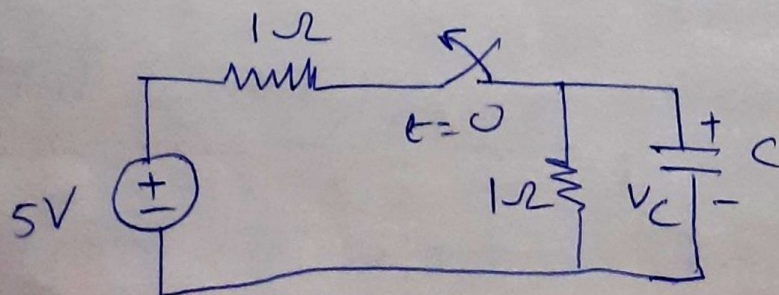
$$i_1(0^-) = 2 - i_L(0^-) = 2 - 1 = 1A$$

current in a resistor can change instantaneously

At  $t = 0^+$ , switch is closed (switch being short ckted)  $\therefore i_1(0^+) = 0A$

$\therefore$  Current in the resistor changes from 1A to 0A

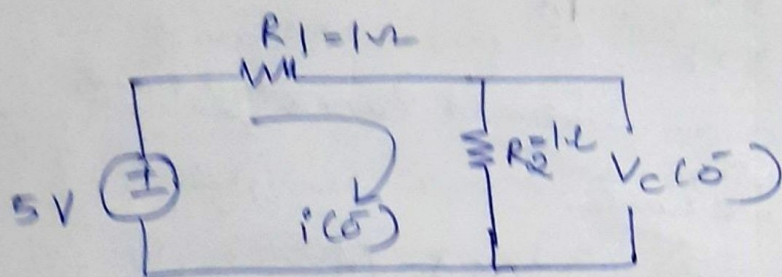
Ex: Refer the ckt, Find  $V_C(0^+)$  Assume that the switch was in closed state for a long time.





Solution: The symbol for the switch implies that it is closed at  $t=0^-$  and then opens at  $t=0^+$

Since the circuit is in steady state with the switch closed, the capacitor is represented as an open circuit at  $t=0^-$



$$V_C(0^-) = R_2 i(0^-)$$

using voltage divider,

$$V_C(0^-) = \frac{V_s}{R_1 + R_2} \cdot R_2 = \frac{5 \times 1}{1 + 1} = 2.5V$$

$$\therefore V_C(0^+) = V_C(0^-) = 2.5V.$$

When the switch is opened at  $t=0$ , if the source is removed from the circuit, still  $V_C(0^-) = 2.5V$ .



### Important note

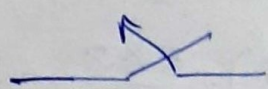
R - current & V/gs changes

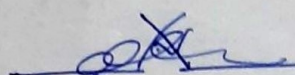
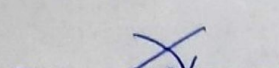
L - current does not change instantaneously

C - V/g does not change instantaneously

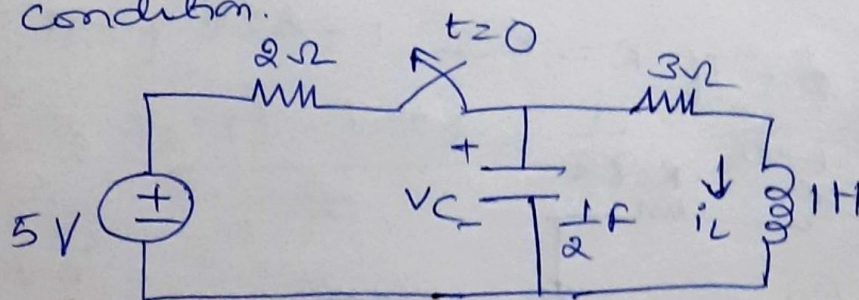
$$\therefore i_L(0^-) = i_L(0^+)$$

$$V_C(0^-) = V_C(0^+)$$

  $\Rightarrow$  closed at  $t=0^-$   
open at  $t=0^+$ ,  $t>0$

  
  $\Rightarrow$  closed at  $t=0^+$ ,  $t>0$   
open at  $t=0^-$

Ex Refer the ckt. Find  $i_L(0^+)$  and  $V_C(0^+)$ . The circuit is in steady state with the switch in closed condition.

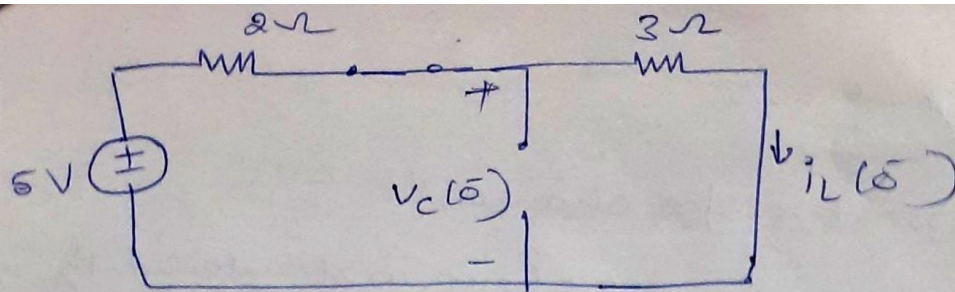


Solution: Switch implies  $\Rightarrow$  closed at  $t=0^-$   
open at  $t=0^+$

To find  $V_C(0^-)$ ,  $i_L(0^-) \Rightarrow$  switch in closed condition  
in steady state

capacitor act as open ckt & inductor act as  
short ckt.





$$i_L(0^-) = \frac{5}{2+3} = 1 \text{ A}$$

$$2 i_L(0^-) + V_C(0^-) = 5$$

$$2 \times 1 + V_C(0^-) = 5$$

$$V_C(0^-) = 5 - 2 = 3 \text{ V}$$

② using vlg divider principle

$$V_C(0^-) = \frac{5 \times 3}{3+2} = \frac{15}{5} = 3 \text{ V}$$

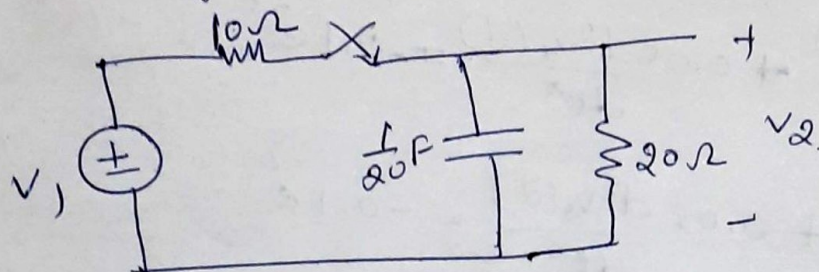
$$\therefore \tilde{i}_L(0^-) = i_L(0^+) = 1 \text{ A}$$

$$V_C(0^-) = V_C(0^+) = 3 \text{ V}$$

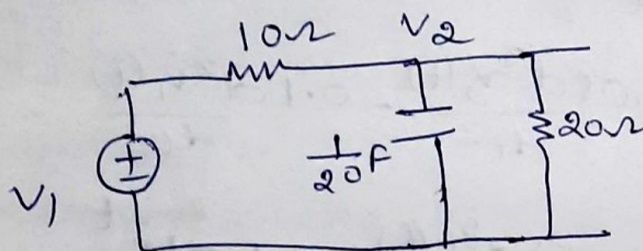


Ex: In the circuit shown in Fig  $V_1(t) = e^{-t}$  for  $t \geq 0$

Determine  $\frac{d^2 V_2}{dt^2}$ ,  $\frac{d^3 V_2}{dt^3}$ , at  $t = 0^+$ . If the capacitor is initially uncharged.



Solution: SW is closed at  $t = 0^+$   
 ——— opened at  $t = 0^-$



$$V_C(0^-) = V_C(0^+) = 0 = V_2(0^+)$$

Apply KCL,

$$\frac{V_2(t) - V_1(t)}{10} + \frac{1}{20} \frac{dV_2(t)}{dt} + \frac{V_2(t)}{20} = 0$$

$t = 0^+$

$$\frac{V_2(0^+) - V_1(0^+)}{10} + \frac{1}{20} \frac{dV_2(0^+)}{dt} + \frac{V_2(0^+)}{20} = 0$$

$$\left(\frac{1}{10} + \frac{1}{20}\right) V_2(t) + \frac{1}{20} \frac{dV_2(t)}{dt} = \frac{V_1(t)}{10} \quad \text{--- (1)}$$

$$0.15 V_2(0^+) + 0.05 \frac{dV_2(0^+)}{dt} = 0.1 e^{-t}$$

$$\therefore V_2(0^+) = 0, \quad \frac{dV_2(0^+)}{dt} = \frac{0.1 e^{-t}}{0.05} = \frac{0.1 e^0}{0.05} = 2 \text{ V/sec}$$



Differentiating equ<sup>n</sup> ①

$$0.15 \frac{dV_2(t)}{dt} + 0.05 \frac{d^2V_2(t)}{dt^2} = 0.1 \frac{dV_1(t)}{dt} \Rightarrow \frac{d\bar{e}^{-t}}{dt} = -\bar{e}^{-t} \quad \text{--- ②}$$

$$t=0^+$$

$$0.15 \frac{dV_2(0^+)}{dt} + 0.05 \frac{d^2V_2(0^+)}{dt^2} = -0.1 \bar{e}^{-t}$$

$$0.15(2) + 0.05 \frac{d^2V_2(0^+)}{dt^2} = -0.1 \bar{e}^0$$

$$\frac{d^2V_2(0^+)}{dt^2} = \frac{-0.1 - 0.3}{0.05} = \underline{\underline{-8 \text{ V/sec}^2}}$$

Differentiating equ<sup>n</sup> ②

$$0.15 \frac{d^2V_2(t)}{dt^2} + 0.05 \frac{d^3V_2(t)}{dt^3} = 0.1 \frac{d^2V_1(t)}{dt^2}$$

$$0.15 \frac{d^2V_2(t)}{dt^2} + (0.05) \frac{d^3V_2(t)}{dt^3} = 0.1 \frac{d^2\bar{e}^{-t}}{dt^2} \Rightarrow t=0^+$$

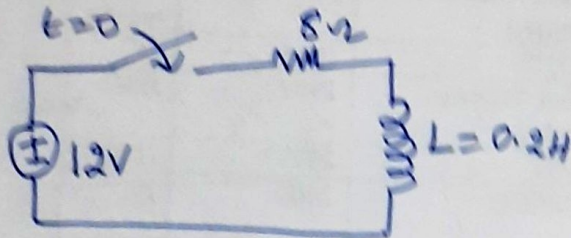
$$(0.15)(-8) + 0.05 \frac{d^3V_2(0^+)}{dt^3} = 0.1(1)$$

$$\frac{d^3V_2(0^+)}{dt^3} = \frac{0.1 + 1.2}{0.05} = 26 \text{ V/sec}^3$$



### Initial Conditions

- ② For the given circuit  $K$  is closed at  $t=0$  with zero current in the inductor. Find the values  $i$ ,  $\frac{di}{dt}$ ,  $\frac{d^2i}{dt^2}$  at  $t=0^+$  if  $R=8\Omega$ ,  $L=0.2H$



given:

Zero current in the inductor

$$\therefore i_L(0^-) = i_L(0^+) = 0$$

$$\begin{array}{l} t \geq 0 \rightarrow t = 0^+ \rightarrow \text{SW is closed} \\ t < 0 \rightarrow t = 0^- \rightarrow \text{" " " " " opened} \end{array} \Rightarrow i(0^+) - i(0^-) = 0$$

Apply KVL,

$$8i + L \frac{di}{dt} = 12 \quad \text{--- (1)}$$

$$\overset{\text{zero}}{8i(0^+)} + 0.2 \frac{di(0^+)}{dt} = 12$$

$$\frac{di(0^+)}{dt} = \frac{12}{0.2} = \underline{\underline{60 \text{ A/sec}}}$$

Differentiating eqn (1)

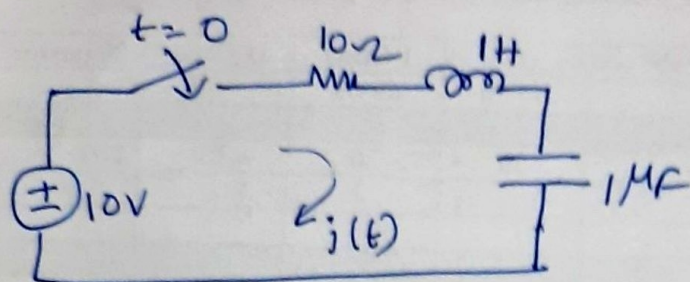
$$8 \frac{di}{dt} + L \frac{d^2i}{dt^2} = 0$$

$$8 \frac{di(0^+)}{dt} + 0.2 \frac{d^2i(0^+)}{dt^2} = 0$$

$$\frac{d^2i(0^+)}{dt^2} = -\frac{8(60)}{0.2} = \underline{\underline{-2400 \text{ A/s}^2}}$$



- ④ In the circuit shown in fig the switch is closed at  $t=0$ , Determine  $i, \frac{di}{dt}, \frac{d^2i}{dt^2}$  at  $t=0^+$



$t > 0 \Rightarrow$  sw is closed  
 $= 0^+$   
 $t < 0 \Rightarrow$  sw is opened

$$10i(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int_0^t i(t) dt = 10$$

Since there is no current through the inductor

at  $t=0^- \Rightarrow i(0^-) = i(0) = 0 \rightarrow \because L$  acts as open

$V_C(0^-) = V_C(0) = 0 \rightarrow \because C$  acts as short

$$10i(t) + \frac{di(t)}{dt} + \frac{1}{1\mu} \int_0^t i(t) dz = 10 \quad \text{--- (1)}$$

$t=0^+$       zero  $\because V_C(0)=0$

$$10i(0^+) + \frac{di(0^+)}{dt} + \frac{1}{1\mu} \int_0^t i(t) dz = 10$$

$$\frac{di(0^+)}{dt} = \underline{\underline{10 \text{ A/sec}}}$$

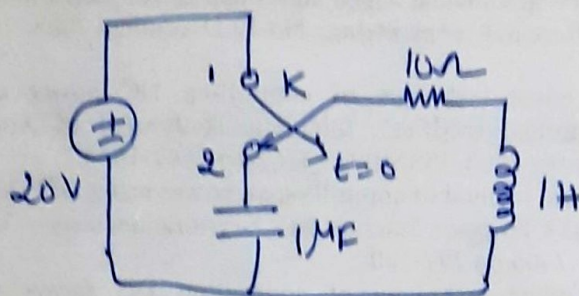
Diff - eqn (1)

$$10 \frac{di(t)}{dt} + \frac{d^2i(t)}{dt^2} + \frac{i(t)}{C} = 0 \quad \text{zero}$$

$$\frac{d^2i(0^+)}{dt^2} = -(10) \times 10 = -100 \text{ A/sec}^2$$



- ⑤ The CRT shown in the fig. The switch  $K$  is (4)  
 changed from position 1 to position 2 at  $t=0$   
 Steady state condition having been reached  
 in position 1. ~~At  $t=0^-$~~  - find the  
 values of  $i$ ,  $\frac{di}{dt}$ ,  $\frac{di}{dt}$  at  $t=0^+$



At position 1

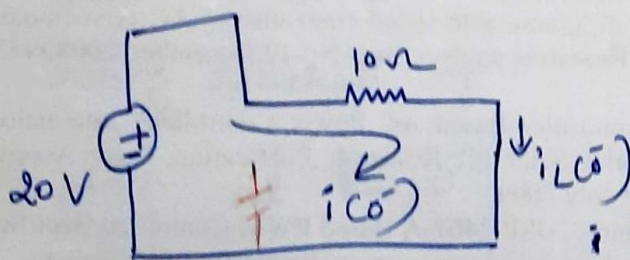
$t=0^-$ ,  $K$  is closed

$t=0^+$ ,  $K$  is opened.

At position 2

$t=0^+$   $K$  is closed

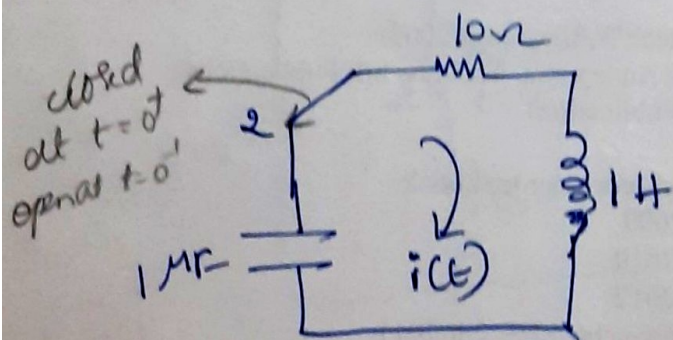
$t=0^-$   $K$  is opened.



$$i_L(0^-) = \frac{20}{10} = 2A$$

$$i(0^-) = i_L(0^-) = i_L(0^+) = 2A$$

$$V_C(0^-) = 0V = V_C(0^+)$$



$$10i(t) + 1 \frac{di(t)}{dt} + \frac{1}{1MF} \int i(t) dt = 0$$

$t=0^+$

$V_C(0^+) = 0$  — (1)

$$10i(0^+) + \frac{di(0^+)}{dt} + 0 = 0$$

$$\frac{di(0^+)}{dt} = -10 \times 2 = -20 A/sec$$



Dyff in equ<sup>n</sup> ①

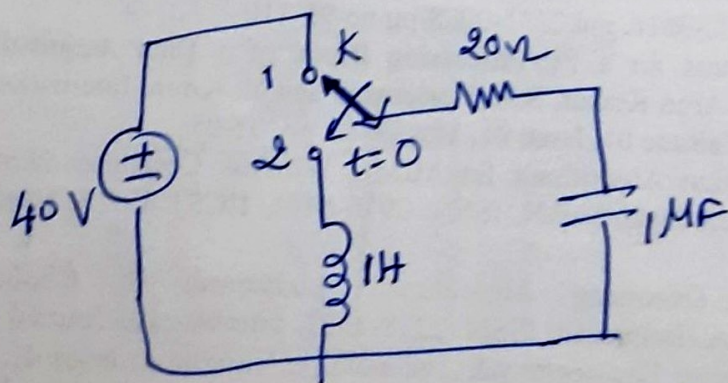
$$10 \frac{di(t)}{dt} + \frac{d^2 i(t)}{dt^2} + \frac{i(t)}{C} = 0$$

$$10 \frac{di(0^+)}{dt} + \frac{d^2 i(0^+)}{dt^2} + \frac{i(0^+)}{1\mu} = 0$$

$$10(-20) + \frac{d^2 i(0^+)}{dt^2} + \frac{2 \times 10^6}{1} = 0$$

$$\begin{aligned} \frac{d^2 i(0^+)}{dt^2} &= 20 - 2 \times 10^6 \\ &= -2 \times 10^6 \text{ A/sec}^2 \end{aligned}$$

⑤ In the n/w of fig, the switch is moved from position 1 to position 2 at  $t=0$ . The steady state having reached before switching calculate  $i$ ,  $\frac{di}{dt}$ ,  $\frac{d^2 i}{dt^2}$  at  $t=0^+$ .



$$V_C(0^-) = V_C(0^+) = 40\text{V}$$

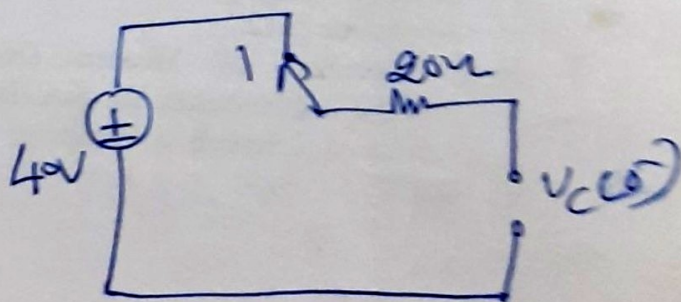
$$i(0^-) = i(0^+) = 0$$

position 1  $\rightarrow$  ~~position 2~~

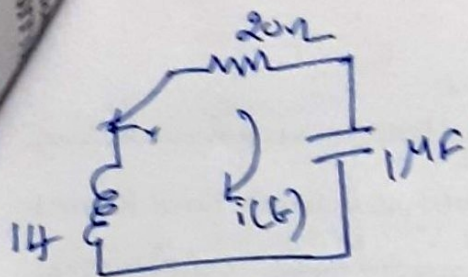
$t=0^-$  SW is closed open

$t=0^+$  SW to ~~open~~ closed

position 2  $\rightarrow$   $t=0^-$  SW is open  
 $t=0^+$  — closed







Apply KVL

$$20i(t) + \frac{1}{C} \int i(z) dz + 1 \frac{di(t)}{dt} = 0$$

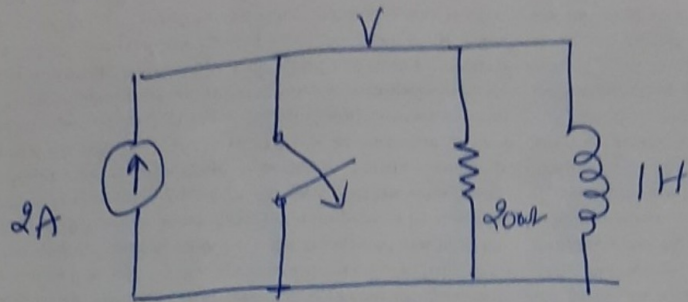
$$20i(t) + v_c(t) + 1 \frac{di(t)}{dt} = 0$$

$$\frac{di(t)}{dt} = -40 \text{ A/sec}$$

$$\frac{d^2i(t)}{dt^2} = 800 \text{ A/sec}^2$$



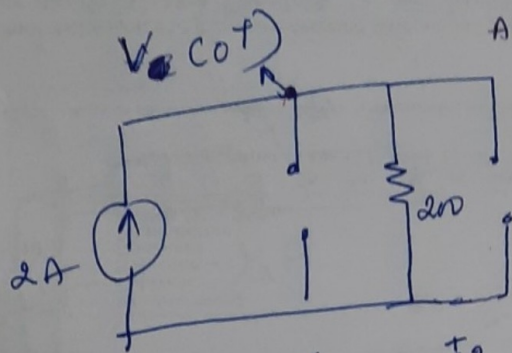
- ⑧ In the given n/w, the switch K is opened at  $t=0$ . At  $t=0^+$  solve for  $V$ ,  $\frac{dV}{dt}$ ,  $\frac{d^2V}{dt^2}$



Solution:

$t=0^- \rightarrow$  SW is closed ( $t < 0$ )  
 $t=0^+ \rightarrow$  SW is opened ( $t > 0$ )

At  $t=0$ , the inductor acts as an open circuit

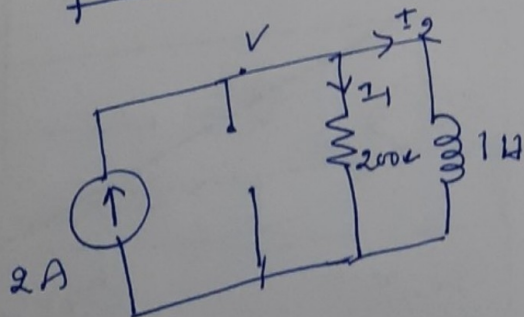


$$V(0^-) = I \cdot R = 2 \times 200 = 400V$$

$$i_L(0^-) = i_L(0^+) = 0A$$

Writing KCL equation

$$I_1 + I_2 - 2 = 0$$



$$\frac{V(t)}{200} + \frac{1}{L} \int V(\tau) d\tau = 2 \quad \text{--- (1)}$$

Diff eqn (1)

$$\frac{1}{L} V(t) + \frac{1}{200} \frac{dV(t)}{dt} = 0 \quad \text{--- (2) } t=0$$

$$\frac{dV(t)}{dt} = \frac{\frac{1}{L} V(0^+)}{\left(\frac{1}{200}\right)} = -\frac{400 \times 200}{1H}$$

$$\frac{dV(0^+)}{dt} = -8 \times 10^4 \text{ V/sec}$$

Diff eqn (2)

$$\frac{1}{L} \frac{dV(t)}{dt} + \frac{1}{200} \frac{d^2V(t)}{dt^2} = 0$$

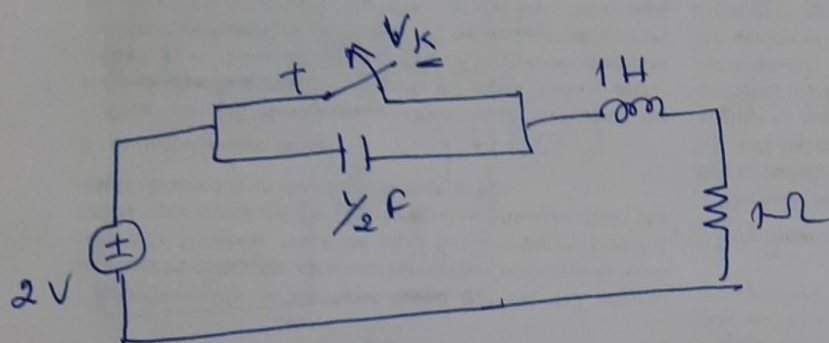


$$-8 \times 10^4 + \frac{1}{2 \times 10^{-6}} \frac{d^2 V(t^+)}{dt^2} = 0$$

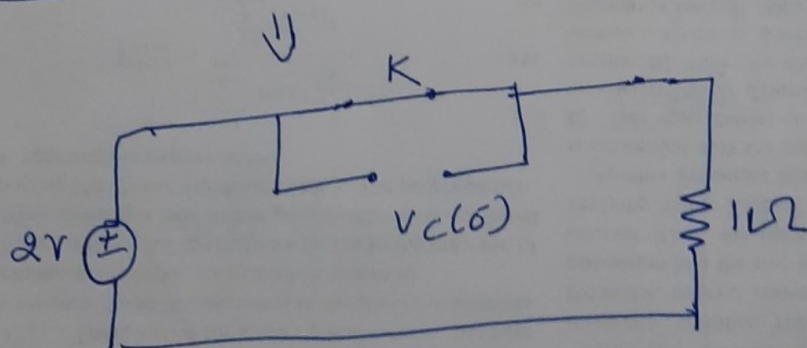
$$\frac{d^2 V(t^+)}{dt^2} = (8 \times 10^4) \times 2 \times 10^{-6}$$

$$= 16 \times 10^6 \text{ V/sec}^2$$

- ⑨ The ckt is in steady state with switch  $K$  closed. At  $t=0$ , the switch is opened. Det the  $V_K$  ac the switch  $V_K$ ,  $\frac{dV_K}{dt}$  at  $t=0^+$



$t < 0$ , SW is closed  
 $t \geq 0$ , SW is opened

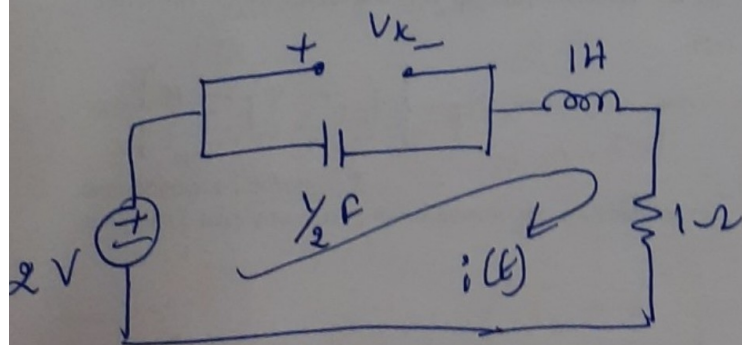


$$i(0^-) = \frac{2}{1}$$

$$= 2 \text{ A} = i(0^+)$$

$$V_C(0^-) = V_C(0^+) = 0$$

$$V_K(0^-) = V_K(0^+) = 0 \text{ V.}$$



$$V_K = \text{voltage}$$

$$i(t) = \text{current}$$

$$i(t) = C \frac{dV_K(t)}{dt}$$

$$\frac{dV_K(0^+)}{dt} = \frac{2 \times 2}{1}$$

$$= 4 \text{ V/sec}$$

$$i(0^+) = \frac{1}{2} \frac{dV_K(0^+)}{dt}$$